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Институт цифровых
интеллектуальных систем

Кафедра
компьютерных систем управления

Дисциплина «Основы системного программного обеспечения»

Отчет по лабораторной работе №1

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Содержание

1. Цель: Освоить работу с системой контроля версий Git и платформой GitHub, включая клонирование репозитория, совместную работу, отправку изменений и разрешение конфликтов.

2. Задачи работы:

1. Создать и настроить локальный репозиторий Git.
2. Освоить основные команды Git: `add`, `commit`, `status`, `push`, `pull`.
3. Научиться создавать и объединять ветки репозитория.
4. Освоить работу с удалённым репозиторием на GitHub.
5. Настроить файл `.gitignore` для игнорирования файлов.
6. Выполнить клонирование репозитория с помощью команды `git clone`.
7. Освоить совместную работу над проектом с использованием GitHub.
8. Выполнить добавление, изменение и отправку файлов в удалённый репозиторий.
9. Получить и разрешить конфликт при совместной работе с репозиторием

3. Описание и выполнение тестовых (если присутствуют) и индивидуальных заданий

Аккаунт на гите уже создан, создаем новый репозиторий, добавляем новую ветку `master`.

Создаем локальную папку на компьютере, создаем текстовый документ
Далее всё по скриншотом, подключаемся к гиту, пушим ветки, коммитим, удаляем, исправляем и наполняем.

```

Windows PowerShell
(C) Корпорация Майкрософт (Microsoft Corporation). Все права защищены.

Установите последнюю версию PowerShell для новых функций и улучшения! https://aka.ms/PSWindows

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git config --global user.name "Max Gorb"
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git config --global user.email maksgorbanev987@gmail.com
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git init
Initialized empty Git repository in C:/Users/maksg/OneDrive/Документы\Хакатон\Gorbanev/.git/
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   test.txt

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "first commit"
[master (root-commit) 5f01f93] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remote add origin https://github.com/Streeelaa/lab1_gorb/tree/master
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remove -v
git: 'remove' is not a git command. See 'git --help'.

The most similar command is
  remote
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remote -v
origin https://github.com/Streeelaa/lab1_gorb/tree/master (fetch)
origin https://github.com/Streeelaa/lab1_gorb/tree/master (push)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git pull origin master
fatal: repository 'https://github.com/Streeelaa/lab1_gorb/tree/master/' not found
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remote rm origin
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remote add origin https://github.com/Streeelaa/lab1_gorb.git
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git remote -v
origin https://github.com/Streeelaa/lab1_gorb.git (fetch)
origin https://github.com/Streeelaa/lab1_gorb.git (push)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git pull origin master
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 855 bytes | 122.00 KiB/s, done.
From https://github.com/Streeelaa/lab1_gorb
 * branch      master      -> FETCH_HEAD
 * [new branch] master      -> origin/master
fatal: refusing to merge unrelated histories
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git pull origin master --allow-unrelated-histories
From https://github.com/Streeelaa/lab1_gorb
 * branch      master      -> FETCH_HEAD
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev>
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 523 bytes | 523.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Streeelaa/lab1_gorb.git
 87ce2e9..e954f8c master -> master
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch
* master
  second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout second
Switched to branch 'second'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> echo Gorbanev > test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add new file to second branch"
[second 9ddd043] add new file to second branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin second
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 344 bytes | 344.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'second' on GitHub by visiting:
remote:   https://github.com/Streeelaa/lab1_gorb/pull/new/second
remote:
To https://github.com/Streeelaa/lab1_gorb.git
 * [new branch]      second -> second
```

second

3 Branches

0 Tags

Switch branches/tags

Find or create a branch...

Branches

Tags

main

master

✓ second

View all branches

README

lab1_gorb

Public

Pin

Watch 0

second

3 Branches

0 Tags

Go to file

T

+

Code

This branch is 3 commits ahead of main .

#1

Streeelaa

add new file to second branch

9ddd043 · 2 minutes ago

4 Commits

README.md	Initial commit	23 minutes ago
test.txt	first commit	16 minutes ago
test_new.txt	add new file to second branch	2 minutes ago

README

lab1_gorb

```

87ce2e9..e954f8c master -> master
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch
* master
  second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout second
Switched to branch 'second'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> echo Gorbanev > test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add new file to second branch"
[second 9ddd043] add new file to second branch
 1 file changed, 0 insertions(+), 0 deletions(-)
   create mode 100644 test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin second
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 344 bytes | 344.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'second' on GitHub by visiting:
remote:   https://github.com/Streeelaa/lab1_gorb/pull/new/second
remote:
To https://github.com/Streeelaa/lab1_gorb.git
 * [new branch]      second -> second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git merge second
Updating e954f8c..9ddd043
Fast-forward
 test_new.txt | Bin 0 -> 22 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
   create mode 100644 test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch -d second
Deleted branch second (was 9ddd043).
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch newdev
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout newdev
Switched to branch 'newdev'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add middle name in newdev"
[newdev 90b809d] add middle name in newdev
 1 file changed, 1 insertion(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add another text in master"
[master 82684f1] add another text in master
 1 file changed, 1 insertion(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```

Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git merge second
Updating e954f8c..9ddd043
Fast-forward
 test_new.txt | Bin 0 -> 22 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test_new.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch -d second
Deleted branch second (was 9ddd043).
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch newdev
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout newdev
Switched to branch 'newdev'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add middle name in newdev"
[newdev 90b809d] add middle name in newdev
 1 file changed, 1 insertion(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add another text in master"
[master 82684f1] add another text in master
 1 file changed, 1 insertion(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Already on 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add changes in master"
[master be3b5a3] add changes in master
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git merge newdev
Auto-merging test.txt
CONFLICT (content): Merge conflict in test.txt
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```

Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git diff
diff --cc test.txt
index d145b4a,09c4167..0000000
--- a/test.txt
+++ b/test.txt
@@@ -1,1 -1,1 +1,5 @@@
-Sergeevich
-+<<<<<<< HEAD
-+ABB - twenty three - null seven da zadolbalo kogda vse
-+=====
-+Sergeevich
-+>>>>>>> newdev
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "test"
[master 6e098da] test
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin master
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.25 KiB | 1.25 MiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Streeelaa/lab1_gorb.git
 e954f8c..6e098da master -> master
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git branch -d newdev
Deleted branch newdev (was 90b809d).
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin --delete newdev git push origin --delete second
error: unable to delete 'newdev': remote ref does not exist
error: unable to delete 'git': remote ref does not exist
error: unable to delete 'push': remote ref does not exist
error: unable to delete 'origin': remote ref does not exist
error: failed to push some refs to 'https://github.com/Streeelaa/lab1_gorb.git'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin --delete newdev
error: unable to delete 'newdev': remote ref does not exist
error: failed to push some refs to 'https://github.com/Streeelaa/lab1_gorb.git'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin --delete second
To https://github.com/Streeelaa/lab1_gorb.git
 - [deleted] second
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```

commit 9ddd04358a227b3e9518350ac63242a92553cfbe
Author: Max Gorb <maksgorbanev987@gmail.com>
Date: Mon May 18 18:39:23 2026 +0300

    add new file to second branch

commit e954f8c6d1de217ba0317f5b6cfe9ef32f32dee8
Merge: 5f01f93 87ce2e9
Author: Max Gorb <maksgorbanev987@gmail.com>
Date: Mon May 18 18:32:49 2026 +0300

    Merge branch 'master' of https://github.com/Streeelaa/lab1_gorb

commit 5f01f93232db32cea5f6173eae6b35081cdd1041
Author: Max Gorb <maksgorbanev987@gmail.com>
Date: Mon May 18 18:25:14 2026 +0300

    first commit

commit 87ce2e9438016e9872c195b857aa59fd910a4828
Author: Streeelaa <maksgorbanev987@gmail.com>
Date: Mon May 18 18:18:52 2026 +0300

    Initial commit
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout 90b809df5c9ed32b24a0246122ec631c88c75bce
Note: switching to '90b809df5c9ed32b24a0246122ec631c88c75bce'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 90b809d add middle name in newdev
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Previous HEAD position was 90b809d add middle name in newdev
Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |

```

```

HEAD is now at 90b809d add middle name in newdev
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git checkout master
Previous HEAD position was 90b809d add middle name in newdev
Switched to branch 'master'
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git reset HEAD
Unstaged changes after reset:
M      test.txt
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "change comment"
[master 5cf1094] change comment
 1 file changed, 2 insertions(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit --amend
[master 2155dcc] alabuga stankin
 Date: Mon May 18 19:07:42 2026 +0300
 1 file changed, 2 insertions(+)
PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git log
commit 2155dcc2803994baab0a74cddceb2090d61e3493 (HEAD -> master)
Author: Max Gorb <maksgorbanev987@gmail.com>
Date: Mon May 18 19:07:42 2026 +0300

    alabuga stankin

```


commit 5f01f93232db32cea5f6173eae6b35081cdd1041

Author: Max Gorb <maksgorbanev987@gmail.com>

Date: Mon May 18 18:25:14 2026 +0300

first commit

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git add .gitignore

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git commit -m "add gitignore"

[master fb65c63] add gitignore

1 file changed, 1 insertion(+)

create mode 100644 .gitignore

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> git push origin master

Enumerating objects: 8, done.

Counting objects: 100% (8/8), done.

Delta compression using up to 16 threads

Compressing objects: 100% (5/5), done.

Writing objects: 100% (6/6), 654 bytes | 654.00 KiB/s, done.

Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (1/1), done.

To https://github.com/Streeelaa/lab1_gorb.git

6e098da..fb65c63 master -> master

PS C:\Users\maksg\OneDrive\Документы\Хакатон\Gorbanev> |